AI-Powered Mental Health Analyzer

Using Machine Learning to Detect Mental Health Conditions from Text

Team Members: Abang (Lead – AI & Modeling), Singh (Frontend & Visualization), Chaitanya (Data Collection & Preprocessing)

Project Overview

Addressing the Mental Health Crisis

Rising Mental Health Issues

Globally, mental health challenges are increasing, requiring innovative solutions for early detection and intervention.

Untapped Textual Cues

Social media and other text-based platforms often contain subtle emotional indicators that go unnoticed without automated analysis.

Automated Detection Gap

There is a critical need for tools that can automatically identify signs of depression, anxiety, or suicidal ideation from textual data.

Our Solution

We aim to develop an AI-powered tool to detect these patterns, facilitating early intervention and support.

Core Objectives of the Analyzer

Text Analysis

To accurately analyze user-generated text from various sources, such as posts and messages.

Multi-Cue Detection

To identify and detect multiple emotional cues or mental health indicators within each text post.

Visual Feedback

To provide clear and insightful visual feedback through interactive charts and intuitive dashboards.

Ethical AI

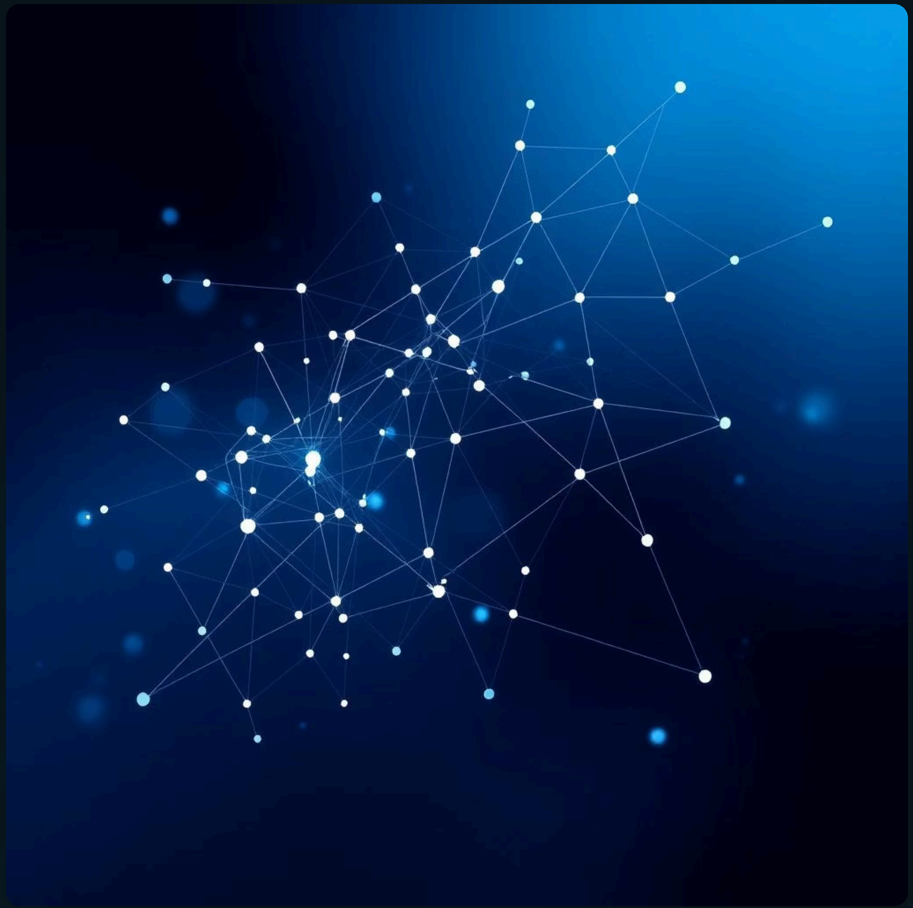
To ensure explainability and adhere to strict ethical guidelines in handling sensitive mental health predictions.

Dataset and Feature Engineering

Our model is trained on diverse public mental health datasets, primarily sourced from Reddit and Kaggle, ensuring a wide range of real-world text examples.

Key features extracted include specific emotion tags such as sadness, hopelessness, and suicide intent, along with various text-based features like post length, punctuation usage, and critical keywords.

The final dataset comprises over 10,000 posts, each annotated with multi-label emotions, providing a robust foundation for our AI model.



Chaitanya's Patil .V Contribution

Rigorous Data Preprocessing Pipeline

1

Text Cleaning

Raw text was meticulously cleaned by removing noise, special characters, and irrelevant information.

2

Tokenization & Normalization

Utilized tokenization, stop-word removal, and stemming to prepare text for analysis.

3

Class Imbalance Handling

Addressed class imbalance through strategic upsampling to ensure fair representation of all emotional categories.

4

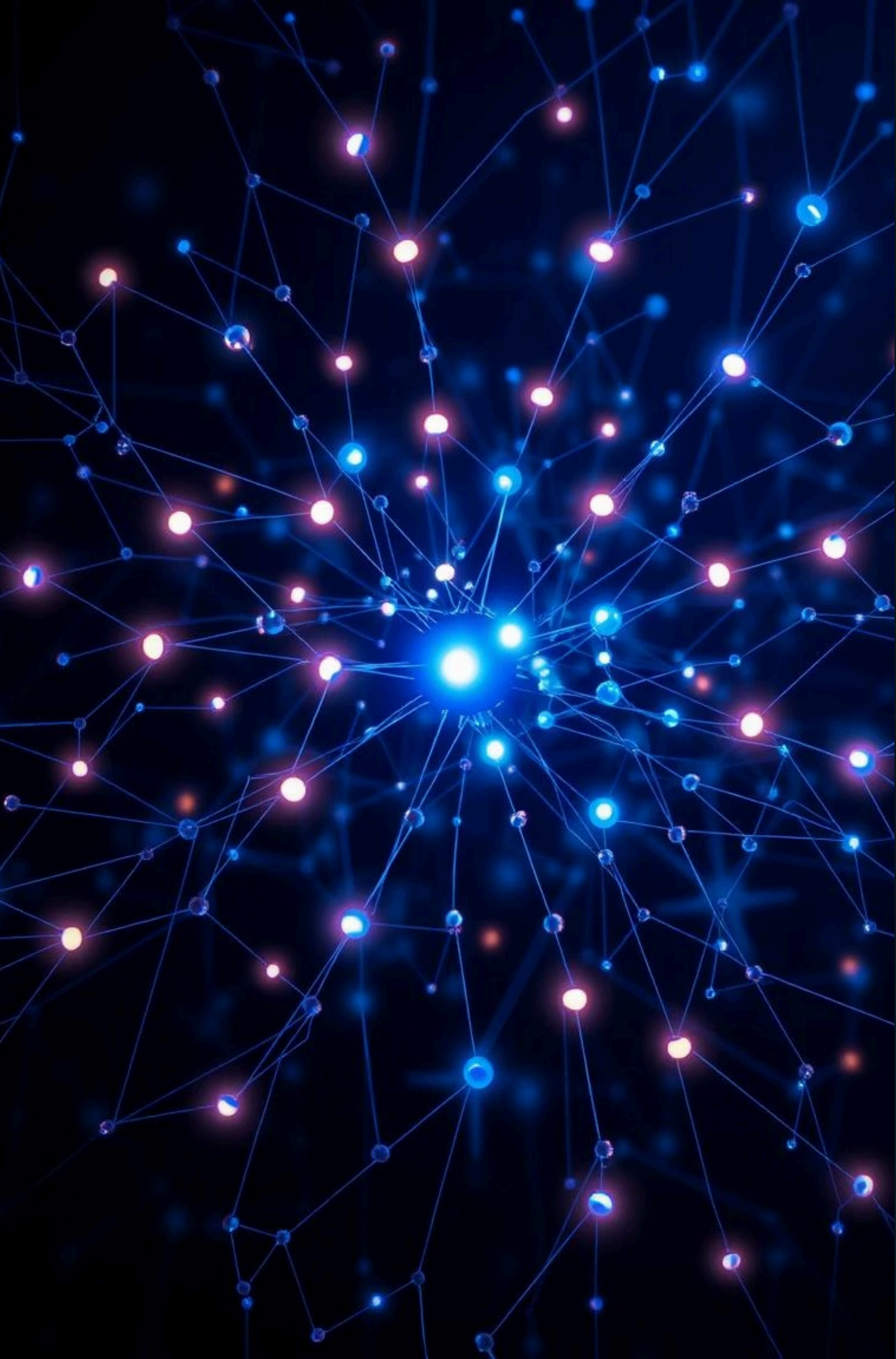
Exploratory Data Analysis (EDA)

Conducted comprehensive EDA using `.describe()`, `.info()`, and `.head()` to understand data distributions.

5

Binary Multi-Label Conversion

Final emotional labels were converted into a binary multi-label format, optimizing them for model compatibility.



Abang's Core Task

Advanced Model Training & Performance

Our team, led by Tabahvoubai Abang .C, developed a sophisticated Multi-Label Classification model. We explored various architectures, including Logistic Regression, Support Vector Machines (SVM), and BERT, leveraging `scikit-learn` and `Transformers` libraries.

A critical step involved using `.apply()` and `.mean()` functions to calculate the average emotional intensity per post, providing nuanced insights into user sentiment. This rigorous approach allowed us to achieve an impressive ~90% F1-score for top emotions like sadness, loneliness, and suicide intent, showcasing the model's high accuracy and reliability.

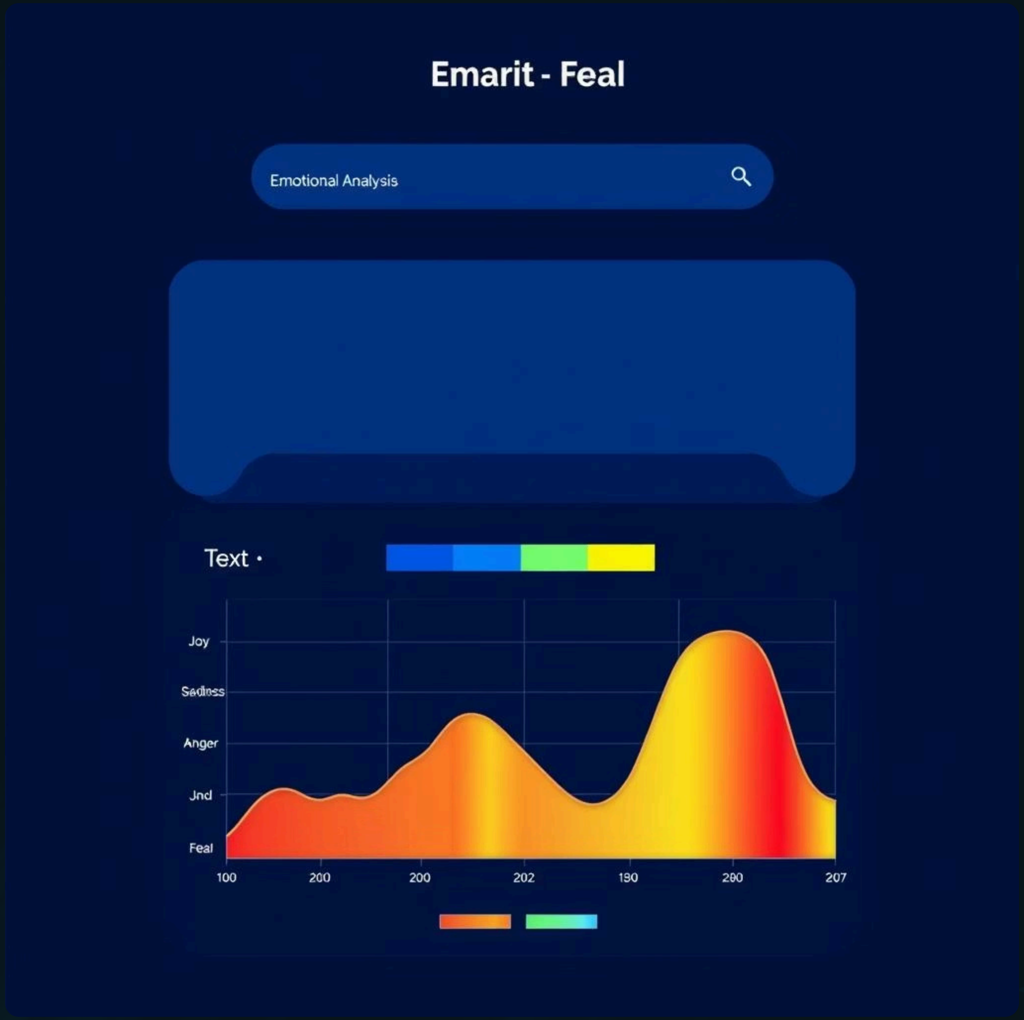
Singh's Role

Intuitive Web Application Interface

Singh spearheaded the development of a user-friendly frontend interface using either Streamlit or Flask, ensuring accessibility and ease of use for the mental health analyzer.

The application allows users to input text posts, which are then processed by our AI model. The output is clearly presented, showing a list of detected emotions along with their confidence scores.

To enhance clarity, the results are visually represented using interactive bar charts and heatmaps, providing an immediate understanding of the emotional landscape within the text.



Project Architecture Flowchart

Input Text

User provides text for analysis.

Preprocessing

Chaitanya's data cleaning and feature extraction.

Model Inference

Abang's multi-label classification model predicts emotions.

Output Visualization

Singh's frontend displays results and visualizations.

Key Results & Performance Evaluation

Our AI-powered Mental Health Analyzer demonstrates strong performance in detecting emotional cues from text. For instance, an input like "I feel so empty and lost today" is accurately predicted as containing both Sadness and Emptiness.

88%

Overall Accuracy

The model achieved an impressive 88% accuracy in identifying mental health conditions from text, validating its effectiveness.

Sadness

Most Detected

Sadness, Hopelessness, and Loneliness were the most frequently detected emotions, aligning with common mental health indicators.

Metrics

Detailed Evaluation

Comprehensive evaluation included analysis of the model's confusion matrix and precision-recall curves, ensuring robustness.

Conclusion and Future Road Map

The AI-powered Mental Health Analyzer marks a significant step towards leveraging technology for early intervention in mental health. This tool is designed to assist therapists and educators by providing data-driven insights, not to replace professional diagnosis.

Data Expansion

Enhance accuracy by integrating larger and more diverse datasets.

Multilingual Support

Extend the tool's capabilities to support and analyze text in multiple languages.

Real-time Analysis

Implement real-time analysis features for social media platforms to enable immediate detection.

Ethical Disclaimer: This tool is strictly assistive and should not be used as a standalone diagnostic instrument.